Group: 6 May 15, 2026

Phase/Milestone 2: Analytics Implementation - Model Building Evaluation

by

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Abstract: This project develops a short-term high-risk prediction framework for predictive policing using jurisdiction-level crime data. The task is formulated as a next-window binary classification problem and evaluated under both 7-day and 3-day settings. A unified preprocessing pipeline is applied to Chicago crime records and external NIBRS data, and four models are compared: a persistence-based baseline, logistic regression, random forest, and LSTM.

The results show that random forest is the most robust model overall, combining strong performance in Chicago 2025 with better generalization to NIBRS. Logistic regression remains competitive in the in-domain setting, while LSTM does not show clear advantages in the current setup. Overall, the 7-day setting is more stable than the 3-day setting. These findings suggest that the framework is useful as a decision-support tool for resource allocation and risk prioritization, but should support rather than replace human judgment.

Code repository (GitHub):
`https://github.com/TIMTIAN66/Team6_IT5006_Predictive_Policing_AY2526Sem2`

1 Problem Definition

1.1 Task Formulation

This project formulates predictive policing as a *next-window high-risk prediction* task at the jurisdiction level. Given the historical crime patterns and contextual information of jurisdiction r up to the current time window t, the objective is to predict whether this jurisdiction will enter a high-risk state in the subsequent window $t+1$.

The study adopts a panel spatio-temporal setting in which each sample corresponds to a jurisdiction–time-window pair. Crime incidents are aggregated spatially at the jurisdiction level and temporally into fixed windows. To examine the effect of temporal granularity, two window settings are considered: a 7-day window and a 3-day window. This transforms the original crime records into a supervised binary classification problem.

Formally, let $X_{r,t}$ denote the feature vector of jurisdiction r at time window t, constructed only from information available at or before t. The prediction target is defined as

$$Y_{r,t+1} = 1\left(C_{r,t+1} \geq \tau_w\right),$$

where $C_{r,t+1}$ is the aggregated crime count in the next time window and τ_w is the threshold under window setting w. The model estimates a risk score

$$\hat{p}_{r,t+1} = f(X_{r,t}),$$

which can be further converted into a binary high-risk prediction.

1.2 Analytical Scope

The objective of this study is to identify jurisdictions with elevated short-term crime risk rather than to forecast the exact number or specific type of future offences. Accordingly, the prediction target is defined on the overall crime level of a jurisdiction in the next period. This formulation is more consistent with practical policing needs, where the priority is often to identify high-risk areas for intervention and resource allocation.

To evaluate both forecasting ability and robustness, the task is examined across multiple domains. Models are trained on historical Chicago data, validated on

future observations from the same city, and further tested on external NIBRS data. Therefore, the problem can be characterized as a jurisdiction-level, panel time-series, binary risk classification task for short-term predictive policing.

2 Data Preparation

2.1 Dataset Construction

This study uses two crime data sources: Chicago crime records and external NIBRS data. The Chicago data are split into a training domain (2015–2024) and an in-domain test set (2025), while the NIBRS data are used for out-of-domain evaluation. NIBRS records from California and Washington are standardized and merged into a unified external test set.

The modeling unit is defined as a *jurisdiction–time-window* observation. Incidents are aggregated spatially at the jurisdiction level and temporally under two parallel settings, namely 7-day and 3-day windows.

2.2 Data Cleaning and Harmonization

A unified preprocessing pipeline is applied across all datasets. Key fields, including event date, jurisdiction identifier, and offence category, are standardized into a common schema, and records with missing core information are removed. Because Chicago and NIBRS use different offence taxonomies, all incidents are mapped into six shared crime groups, while unmappable records are excluded.

To support joint modeling, a consistent `region_id` is constructed so that Chicago districts and NIBRS agencies can be treated as comparable jurisdiction-level units. The post-harmonization crime-group distributions are shown in Appendix A.1, and the final panel dimensions after preprocessing and temporal aggregation are summarized in Appendix A.2.

2.3 Feature Engineering

Feature engineering is conducted on the jurisdiction–window panel using only current and historical information. Crime events are aggregated into window-level

counts, and the full region–time grid is completed so that windows with no incidents are explicitly represented.

The resulting features summarize recent temporal dynamics, neighboring context, crime composition, calendar seasonality, and long-run regional baseline differences. To avoid information leakage, all predictors are constructed from time t or earlier, while region-level baseline statistics are estimated only from the Chicago training set and then applied unchanged to the evaluation domains. A summary of the feature design is provided in Appendix A.3.

The final output of this stage is a unified spatio-temporal modeling table, in which each jurisdiction–time observation is paired with a binary high-risk label for the next time window.

3 Model Implementation

3.1 Model Design

To compare different levels of modeling complexity under a unified framework, this study implements four representative approaches: a persistence-based baseline, logistic regression, random forest, and long short-term memory (LSTM). The baseline model is motivated by the naive forecasting principle, which uses the most recent observation as a direct proxy for the near-future state [4]. Logistic regression is included as a standard linear classifier for binary outcomes [2], random forest is adopted as a nonlinear ensemble model for tabular prediction [1], and LSTM is used to model sequential temporal dependence across consecutive windows [3].

These four models provide a compact comparison across rule-based, linear, nonlinear, and sequential learning paradigms. Since all models are trained and evaluated under the same data split and preprocessing pipeline, performance differences can be attributed mainly to their modeling assumptions and representational capacity.

In terms of model structure, the persistence baseline is a non-trainable rule-based model that predicts next-window high risk directly from the current-window crime level. Logistic regression models the task as linear binary classification, random forest captures nonlinear interactions through ensemble tree learning, and LSTM

models temporal dependence across consecutive windows through ordered input sequences. Despite these architectural differences, all models are designed to solve the same next-window high-risk classification task, ensuring that comparison is conducted under a common prediction target.

3.2 Training Strategy

All models are trained on Chicago data from 2015 to 2024, evaluated in-domain on Chicago 2025, and further tested out-of-domain on the merged NIBRS dataset. The prediction target is defined consistently across models as a binary indicator of whether the total crime count in the next time window exceeds a threshold estimated exclusively from the Chicago training domain. The same threshold is then fixed for all evaluation domains.

All learning-based models use the same feature representation derived from the data preparation pipeline. Each jurisdiction–time-window observation is encoded as a numerical feature vector, with missing values imputed and continuous variables standardized using statistics estimated only from the Chicago training data. These fitted preprocessing parameters are then transferred unchanged to the in-domain and out-of-domain evaluation sets. For logistic regression and random forest, each sample is treated as an independent tabular observation, whereas for LSTM, observations are reorganized into chronological fixed-length sequences for each jurisdiction.

To examine the effect of temporal granularity, the full modeling pipeline is repeated under both 7-day and 3-day window settings. Training is further repeated across multiple random seeds, and final results are reported using the corresponding mean and standard deviation. Throughout this process, preprocessing statistics, label thresholds, and sequence alignment are all restricted to training-domain information in order to prevent information leakage.

The current implementation adopts a fixed-default hyperparameter strategy rather than extensive automated tuning. This design is intended to establish a stable and comparable baseline across model classes before further optimization. The default settings used for logistic regression, random forest, and LSTM are reported in Appendix A.4. Although no exhaustive hyperparameter search is conducted in the current study, the implementation is structured to support future tuning while preserving the same temporal split, leakage control rules, and evaluation protocol.

4 Model Evaluation

4.1 Evaluation Metrics

Model performance is evaluated using four standard binary classification metrics: Precision, Recall, F1-score, and AUC-ROC. Their formal definitions and interpretations are provided in Appendix A.5. These metrics are reported jointly because the prediction task is subject to substantial class imbalance, particularly in the out-of-domain setting.

As shown in Appendix A.6, the positive rate is around 0.25 in the Chicago training domain and around 0.20 in Chicago 2025, but drops to approximately 0.02 in NIBRS under both the 7-day and 3-day settings. Under such conditions, Precision alone may be misleading, since a conservative classifier can achieve high Precision while missing a large proportion of true positives. For this reason, model performance is assessed using multiple complementary metrics rather than any single summary score.

In addition to label imbalance, the evaluation also reflects cross-domain variation between Chicago and NIBRS. Although all incidents are harmonized into the same six crime groups, the post-harmonization distributions remain more similar between Chicago train and Chicago 2025 than between Chicago and NIBRS, as illustrated in Appendix A.1. Therefore, the reported metrics should be interpreted not only as measures of predictive accuracy but also as indicators of robustness under domain shift.

4.2 Overall Predictive Performance

The complete numerical results are reported in Appendix A.8, and the corresponding comparisons of AUC-ROC, F1-score, Precision, and Recall are shown in Appendix A.7.

For Chicago 2025, logistic regression and random forest are the strongest models overall. Under the 7-day setting, random forest achieves the highest F1-score (0.761), while logistic regression attains the highest AUC-ROC (0.954) and Recall (0.907). Under the 3-day setting, logistic regression remains the best model overall, with the highest AUC-ROC (0.932) and F1-score (0.704). These results indicate that the in-domain task is well captured by the tabular feature representa-

tion, especially for the stronger tabular models.

For NIBRS, the ranking changes substantially. Random forest and the persistence baseline clearly outperform logistic regression and LSTM under both temporal settings. In particular, random forest achieves the highest F1-score in both 7-day (0.951) and 3-day (0.928) evaluation, while the persistence baseline remains highly competitive. Logistic regression shows extremely high Precision in NIBRS, but this is accompanied by much lower Recall, indicating an overly conservative prediction pattern under domain shift. LSTM shows the weakest and least stable performance overall, especially in the 3-day out-of-domain setting.

Overall, the results suggest that random forest is the most robust model across domains, logistic regression is mainly competitive in the in-domain setting, and the 7-day setting is generally more stable than the 3-day setting.

4.3 Spatial and Temporal Evaluation

The detailed spatial and temporal results are reported in Appendix A.9 and Appendix A.10. Overall, the spatial results show that the models are more useful for identifying high-risk areas for deployment than for matching the exact region perfectly, while the temporal results show that the 7-day setting is generally more stable than the 3-day setting.

For spatial evaluation, the Top-K coverage results on NIBRS are the most informative because this dataset contains many more jurisdictions than Chicago. Under the 7-day setting, Top-100 coverage reaches 0.648 for the persistence baseline and 0.576 for random forest, compared with only 0.280 for logistic regression. A similar pattern appears under the 3-day setting, where the baseline and random forest again perform better than the other models. In Chicago 2025, Top-K coverage becomes less useful because the total number of regions is small, so the curves saturate quickly.

The geographic hitmap results show that many spatial errors are local. For the best 7-day model, the exact event hit rate is 0.291, but the neighbor-aware hit rate increases to 0.626. Under the 3-day setting, the best exact hit rate is 0.295, while the neighbor-aware hit rate rises to 0.706. This means that even when the model does not predict the exact same jurisdiction, it often predicts a nearby one. Therefore, its spatial performance is more meaningful for practical deployment than for strict exact-region matching.

The temporal results lead to a similar conclusion. In Chicago 2025, logistic regression and random forest both achieve stronger monthly F1 performance under the 7-day setting than under the 3-day setting. In NIBRS, random forest and the persistence baseline remain the most stable models over time, while LSTM shows much weaker and more volatile monthly performance. Overall, these results suggest that the main predictive signal is reasonably stable over time, but spatial accuracy should be interpreted mainly in terms of operational usefulness rather than exact geographic matching.

4.4 Robustness and Cross-window Comparison

Overall, the results are reasonably consistent across random seeds, and the main ranking of models remains stable under both time-window settings. Random forest shows the strongest robustness across domains, maintaining high performance in both Chicago 2025 and NIBRS, while logistic regression is more sensitive to domain shift and LSTM exhibits the largest instability, especially in the 3-day setting. Comparing the two temporal resolutions, the 7-day setting generally provides better stability and a more balanced Precision–Recall trade-off than the 3-day setting. Therefore, from both a performance and robustness perspective, random forest under the 7-day setting is the most reliable overall configuration for this study.

5 Key Findings and Practical Implications

The results show that recent crime patterns are the most useful signals for predicting short-term risk. In particular, the current crime level, recent changes in crime activity, and the long-term baseline of each region play the biggest role in model performance. Logistic regression depends more strongly on recent trend features, while random forest makes use of a wider range of information, including historical patterns, neighboring context, crime structure, and seasonal signals. This helps explain why random forest performs best overall and remains more stable across different domains. By contrast, LSTM does not provide clear additional benefits in the current setting, especially when the model is tested on external data.

From a practical point of view, the findings suggest that the model is most useful as a decision-support tool for resource allocation rather than as an exact district-matching system. Some districts appear repeatedly as high-risk areas across both the 7-day and 3-day analyses, which means they should receive more consistent

attention in patrol planning. The results also show a clear seasonal pattern, with higher risk from May to October and especially from June to September. Overall, the 7-day setting is more suitable for stable weekly planning, while the 3-day setting is better for short-term adjustments. Therefore, the model is best used to help prioritize areas and timing of police deployment, rather than to make fully automated decisions.

6 Limitations

This study still has several limitations. First, the model is built at the jurisdiction level, so it cannot capture finer spatial patterns such as street-level hotspots. Second, the out-of-domain evaluation only uses the merged NIBRS data from California and Washington, which means the generalization results should not be treated as representative of all external settings. Third, in order to align Chicago and NIBRS, the data are simplified into six shared crime groups, which causes some information loss. In addition, the label threshold is estimated only from the Chicago training data, so the much lower positive rate in NIBRS makes cross-domain comparison more difficult. The current feature set is also limited to historical crime, calendar, and spatial proxy variables, without external factors such as weather, population mobility, or socioeconomic conditions. Finally, the model is evaluated mainly in terms of predictive performance, but issues such as probability calibration, fairness, and feedback-loop risk still need further study before real deployment.

7 Conclusion

This project develops a complete pipeline for short-term high-risk prediction in predictive policing, covering data harmonization, feature construction, model training, and cross-domain evaluation. The results show that random forest is the most reliable model overall, especially under the 7-day setting, because it combines strong in-domain performance with better robustness on the external NIBRS data. Logistic regression remains competitive in Chicago 2025, while LSTM does not provide clear advantages in the current setting. More broadly, the findings suggest that this framework is useful as a decision-support tool for prioritizing areas and time periods with higher risk, but it should not be treated as a fully automatic decision system. In practice, the model is better used to support human judg-

ment, resource allocation, and early warning rather than to replace operational decision-making directly.

References

[1] L. Breiman. Random forests. *Machine learning*, 45(1):5–32, 2001.

[2] D. R. Cox. The regression analysis of binary sequences. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 20(2):215–232, 1958.

[3] S. Hochreiter and J. Schmidhuber. Long short-term memory. *Neural computation*, 9(8):1735–1780, 1997.

[4] R. J. Hyndman and G. Athanasopoulos. *Forecasting: principles and practice*. OTexts, 2018.

A Supplementary Data Summaries

A.1 Post-harmonization Crime Group Distributions

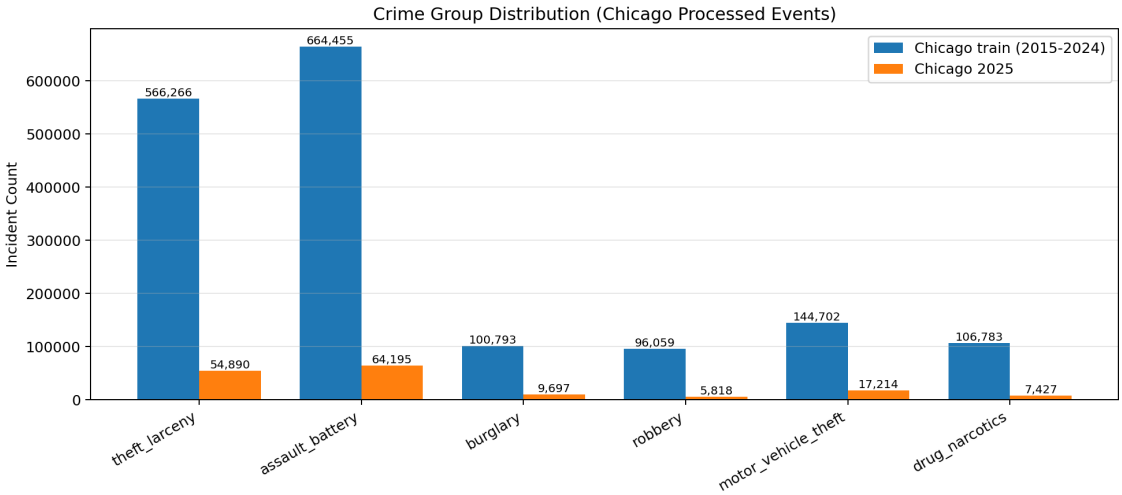

Figure 1: Crime-group distribution of the processed Chicago events after harmonization into the six shared crime categories. The figure compares the Chicago training domain (2015–2024) with the in-domain test set (2025).

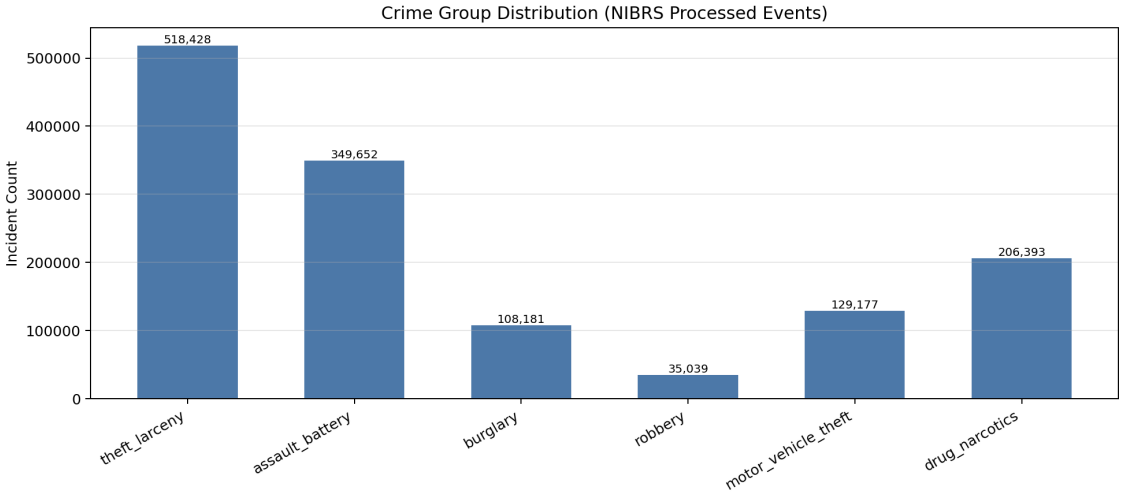

Figure 2: Crime-group distribution of the processed NIBRS events after harmonization into the six shared crime categories. The external evaluation set combines standardized records from California and Washington.

A.2 Final Panel Dimensions After Preprocessing

Table 1: Final panel dimensions after preprocessing, spatial harmonization, and temporal aggregation under the 7-day and 3-day window settings.

Time Window	Dataset	Final Rows	# Regions	# Time Windows
7-day	Chicago train (2015–2024)	12,408	24	517
7-day	Chicago 2025	1,248	24	52
7-day	NIBRS (CA2024+WA2024)	40,368	841	48
3-day	Chicago train (2015–2024)	29,088	24	1,212
3-day	Chicago 2025	2,928	24	122
3-day	NIBRS (CA2024+WA2024)	99,238	841	118

Each row corresponds to a jurisdiction–time-window observation in the final modeling panel after data cleaning, crime-group harmonization, and temporal aggregation.

A.3 Feature Engineering Strategy Summary

Table 2: Summary of the feature engineering strategy, including construction logic, analytical role, leakage control, and cross-domain compatibility.

Feature Block	How It Is Built	Why It Matters	Leakage Control	Cross-domain Compatibility
Temporal windows	Parallel 7-day and 3-day windows; fixed anchor for 3-day binning	Enables comparison of stability and sensitivity across temporal resolutions	Time index is constructed before label shifting; no future alignment is used	Yes
Autoregressive history	Current and lagged counts together with short rolling trend signals	Captures crime persistence and near-term escalation patterns	Uses only current and past windows to predict the next window	Yes
Neighbor-aware context	Same-state peer-region activity summaries and self-versus-neighbor ratios	Adds contextual information beyond single-region history	Neighbor statistics are computed from the same prediction-time slice only	Yes
Crime composition	Last-window shares of the six harmonized crime groups	Encodes structural changes in the local crime mix	Composition features are based on time t, while the target is defined at $t+1$	Yes
Calendar seasonality	Month, week-of-year, quarter, and cyclic temporal encoding	Captures recurring seasonal and periodic crime patterns	Deterministic functions of timestamps with no label information	Yes
Region baseline statistics	Region-level long-run mean, standard deviation, and count estimated from Chicago train only	Provides structural baseline risk and regional variability information	Fitted only on Chicago training data and transferred unchanged to evaluation domains	Yes
Label strategy	Binary high-risk label derived from next-window total count threshold	Produces a policy-oriented alerting target for intervention prioritization	Threshold is estimated only on Chicago training data and fixed for all evaluation domains	Yes

All feature blocks are constructed under a strict temporal-causality constraint, using only information available up to time.

A.4 Default Hyperparameters in the Current Implementation

Table 3: Default hyperparameter settings used in the current implementation for each learning-based model.

Model	Hyperparameter	Default Value
Logistic Regression	learning rate	0.01
Logistic Regression	epochs	220
Logistic Regression	batch size	256
Random Forest	n_estimators	120
Random Forest	max_depth	10
Random Forest	min_samples_split	8
Random Forest	min_samples_leaf	3
Random Forest	max_features	`sqrt(n_features)`
LSTM	sequence length	8
LSTM	hidden_dim	64
LSTM	num_layers	1
LSTM	dropout	0.0
LSTM	learning rate	0.001
LSTM	epochs	80
LSTM	batch size	128

These values are fixed defaults used for the current report and are not the result of exhaustive hyperparameter optimization.

A.5 Definitions of Evaluation Metrics

Table 4: Formulas of the evaluation metrics used in this study.

Metric	**Formula**
Precision	$Precision = \frac{TP}{TP + FP}$
Recall	$Recall = \frac{TP}{TP + FN}$
F1-score	$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$
AUC-ROC	$AUC = \int_0^1 TPR(FPR)\, d(FPR)$

TP, FP, and FN denote true positives, false positives, and false negatives, respectively. TPR and FPR denote true positive rate and false positive rate.

Precision: Measures the proportion of predicted high-risk samples that are truly high-risk. A higher Precision indicates fewer false positives among positive predictions.

Recall: Measures the proportion of true high-risk samples that are successfully identified by the model. A higher Recall indicates fewer false negatives.

F1-score: The harmonic mean of Precision and Recall. It provides a balanced summary when both false positives and false negatives are important.

AUC-ROC: Measures the model's ranking ability across all decision thresholds. A higher AUC-ROC indicates better discrimination between positive and negative samples independent of a fixed threshold.

A.6 Positive Rates Across Domains

Table 5: Positive rates of the binary high-risk label across domains under the 7-day and 3-day window settings. A sample is labeled positive if the total crime count in the next time window exceeds the predefined threshold, and negative otherwise.

Time Window	Dataset	Positive Rate
7-day	Chicago train (2015–2024)	0.2513
7-day	Chicago 2025	0.1955
7-day	NIBRS (CA2024+WA2024)	0.0222
3-day	Chicago train (2015–2024)	0.2537
3-day	Chicago 2025	0.2056
3-day	NIBRS (CA2024+WA2024)	0.0230

Positive rate is defined at the jurisdiction–time-window level rather than at the crime-group level. The same dataset-level positive rate therefore applies to all harmonized crime groups within a given domain and window setting.

A.7 Overall Predictive Performance Figures

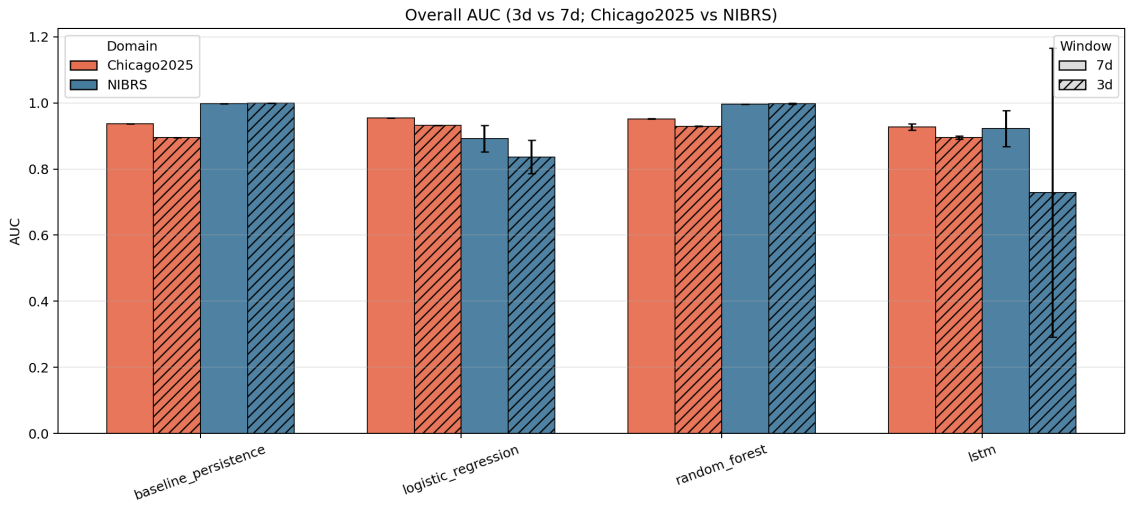

Figure 3: Overall AUC-ROC comparison across models under the 7-day and 3-day settings for Chicago2025 and NIBRS.

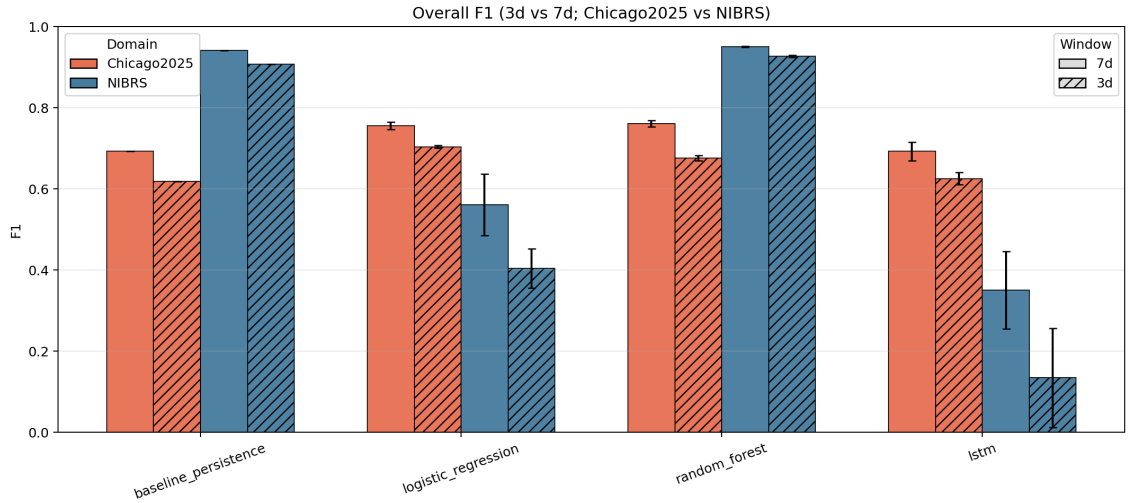

Figure 4: Overall F1-score comparison across models under the 7-day and 3-day settings for Chicago2025 and NIBRS.

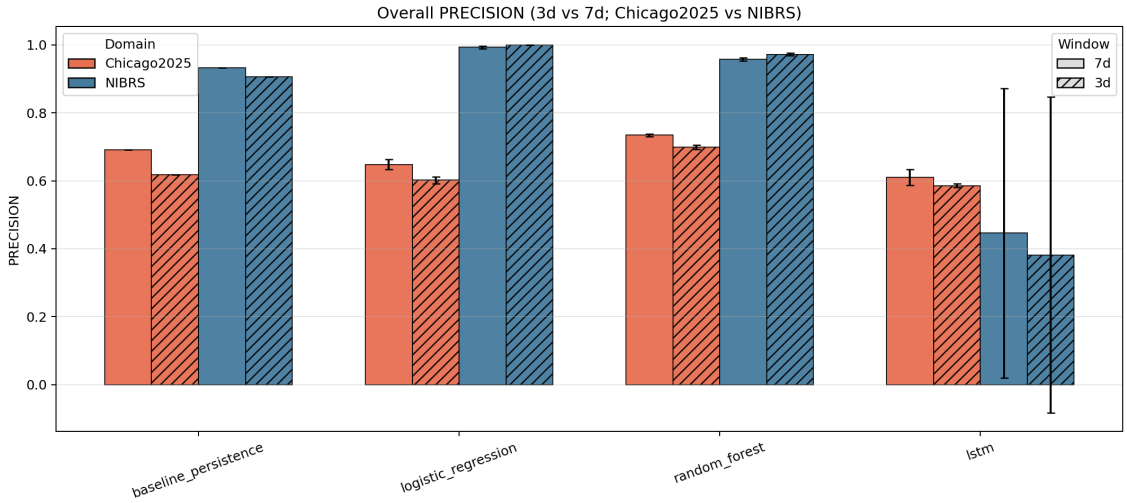

Figure 5: Overall Precision comparison across models under the 7-day and 3-day settings for Chicago2025 and NIBRS.

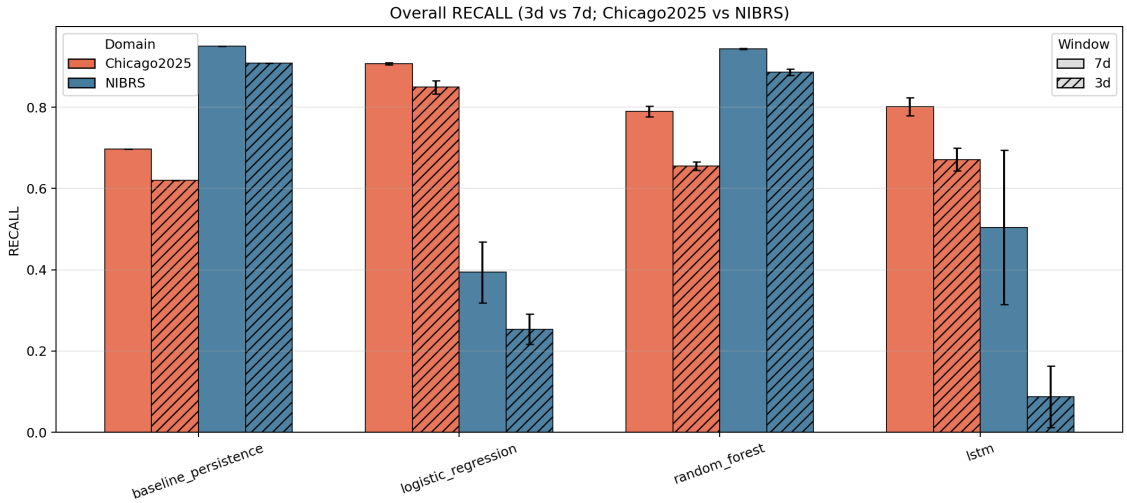

Figure 6: Overall Recall comparison across models under the 7-day and 3-day settings for Chicago2025 and NIBRS.

A.8 Overall Results Across Domains and Time Windows

Table 6: Overall predictive performance across domains and time-window settings. Results are reported as mean ± standard deviation across random seeds.

Time Window	Domain	Model	AUC (mean ± std)	F1 (mean ± std)	Precision (mean ± std)	Recall (mean ± std)
7d	chicago_2025	baseline_persistence	0.937 ± 0.000	0.694 ± 0.000	0.691 ± 0.000	0.697 ± 0.000
7d	chicago_2025	logistic_regression	0.954 ± 0.000	0.756 ± 0.010	0.648 ± 0.015	0.907 ± 0.002
7d	chicago_2025	random_forest	0.952 ± 0.001	0.761 ± 0.007	0.734 ± 0.004	0.790 ± 0.013
7d	chicago_2025	lstm	0.927 ± 0.010	0.693 ± 0.023	0.610 ± 0.024	0.802 ± 0.023
7d	nibrs	baseline_persistence	0.998 ± 0.000	0.942 ± 0.000	0.933 ± 0.000	0.951 ± 0.000
7d	nibrs	logistic_regression	0.892 ± 0.040	0.561 ± 0.075	0.993 ± 0.004	0.394 ± 0.075
7d	nibrs	random_forest	0.997 ± 0.000	0.951 ± 0.002	0.957 ± 0.004	0.944 ± 0.001
7d	nibrs	lstm	0.923 ± 0.055	0.351 ± 0.095	0.446 ± 0.427	0.505 ± 0.190
3d	chicago_2025	baseline_persistence	0.895 ± 0.000	0.619 ± 0.000	0.618 ± 0.000	0.620 ± 0.000
3d	chicago_2025	logistic_regression	0.932 ± 0.000	0.704 ± 0.003	0.602 ± 0.011	0.849 ± 0.017
3d	chicago_2025	random_forest	0.930 ± 0.000	0.677 ± 0.007	0.699 ± 0.007	0.656 ± 0.011
3d	chicago_2025	lstm	0.895 ± 0.005	0.626 ± 0.015	0.586 ± 0.006	0.672 ± 0.028
3d	nibrs	baseline_persistence	0.999 ± 0.000	0.907 ± 0.000	0.906 ± 0.000	0.909 ± 0.000
3d	nibrs	logistic_regression	0.837 ± 0.051	0.404 ± 0.048	1.000 ± 0.000	0.254 ± 0.038
3d	nibrs	random_forest	0.998 ± 0.001	0.928 ± 0.002	0.973 ± 0.004	0.887 ± 0.008
3d	nibrs	lstm	0.729 ± 0.438	0.135 ± 0.122	0.381 ± 0.465	0.087 ± 0.076

Results are aggregated across repeated runs where applicable. Standard deviations reflect variability across random s

A.9 Spatial Evaluation Figures

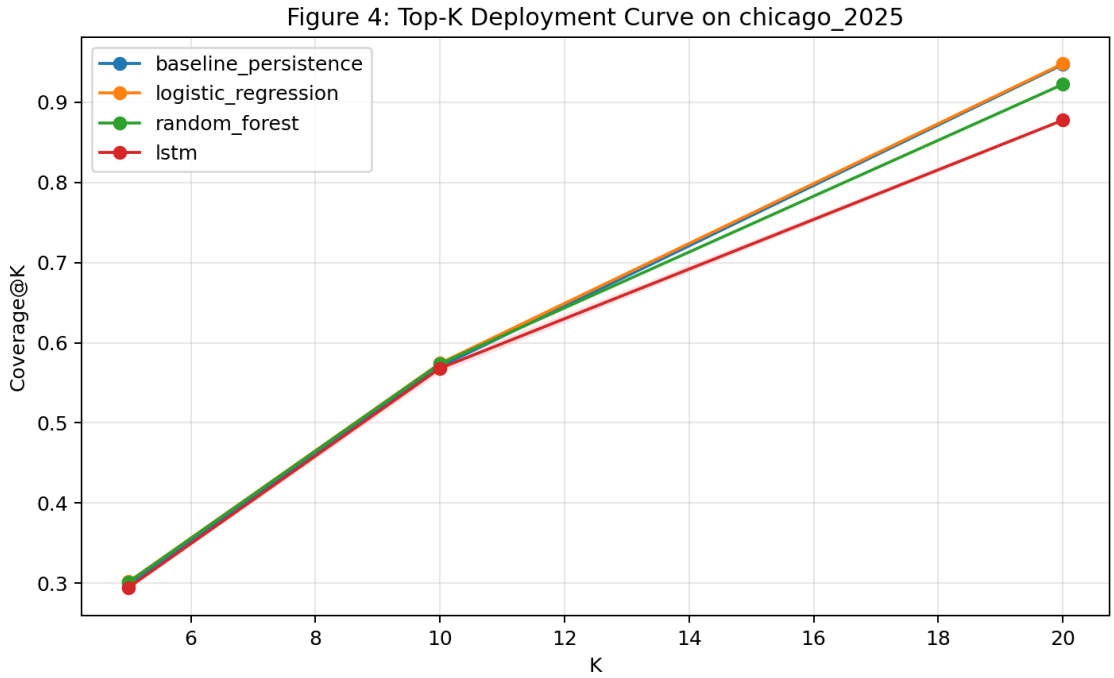

Figure 7: Top-K coverage in Chicago 2025 under the 7-day setting.

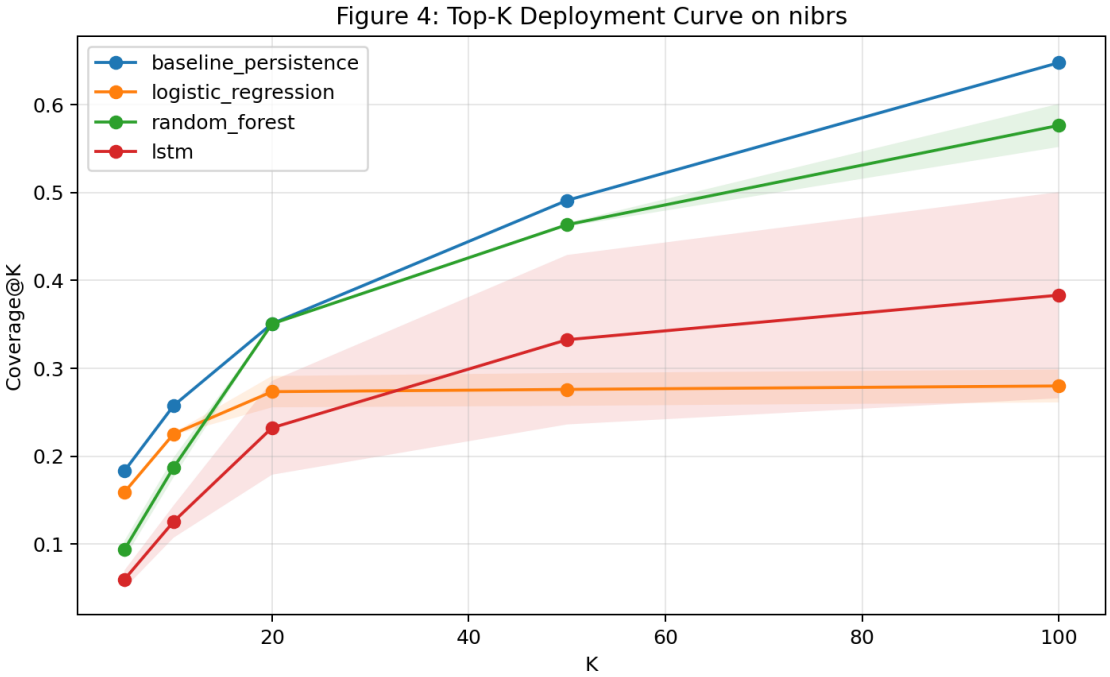

Figure 8: Top-K coverage in NIBRS under the 7-day setting.

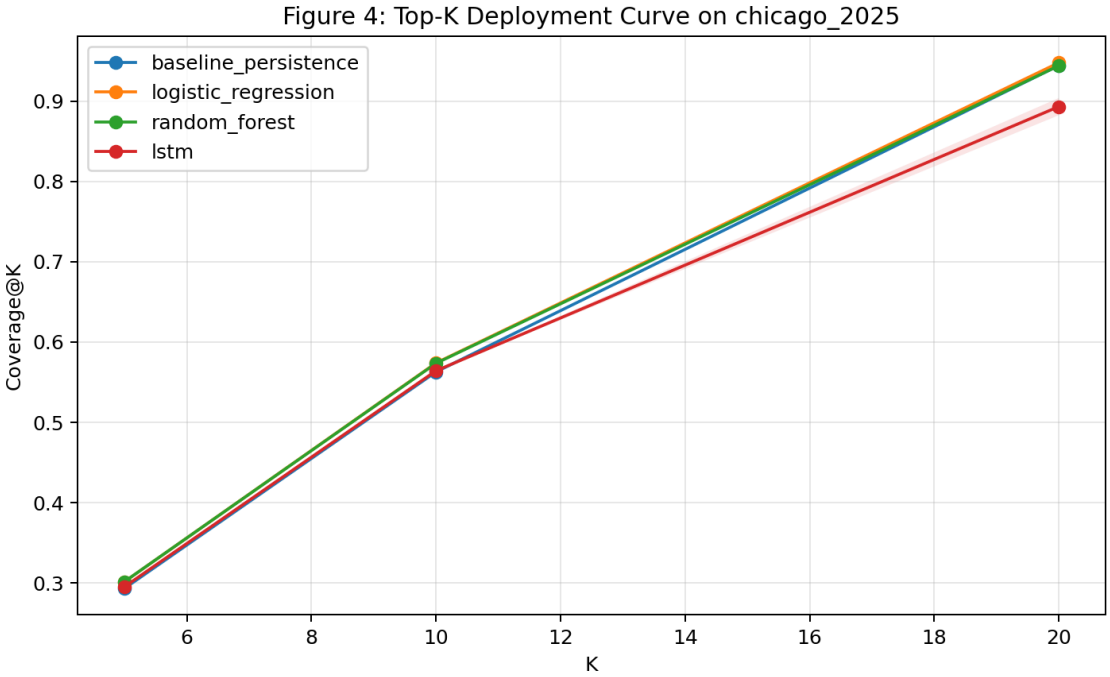

Figure 9: Top-K coverage in Chicago 2025 under the 3-day setting.

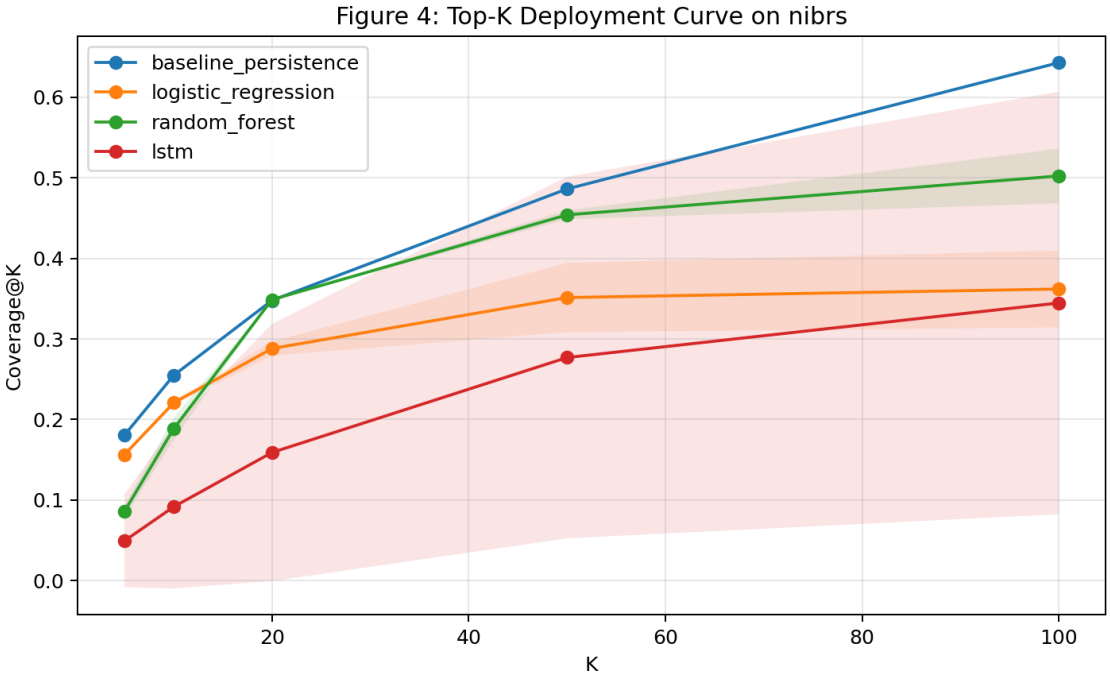

Figure 10: Top-K coverage in NIBRS under the 3-day setting.

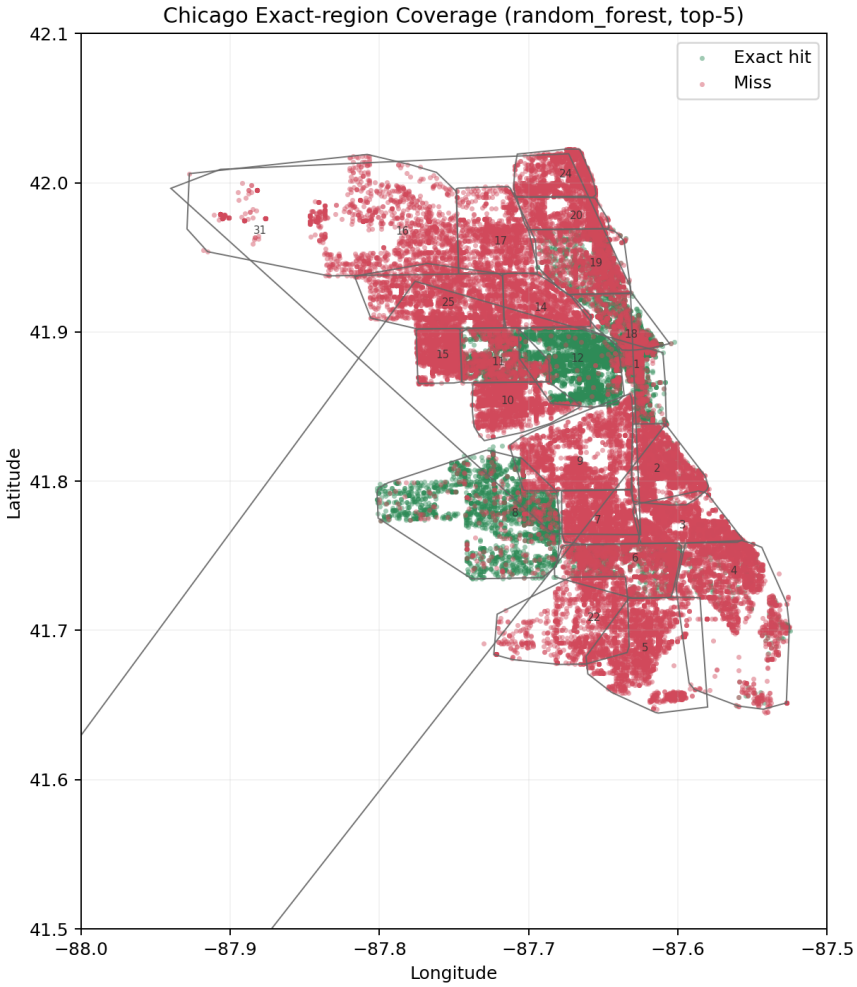

Figure 11: Exact geographic event hitmap for Chicago under the 7-day setting.

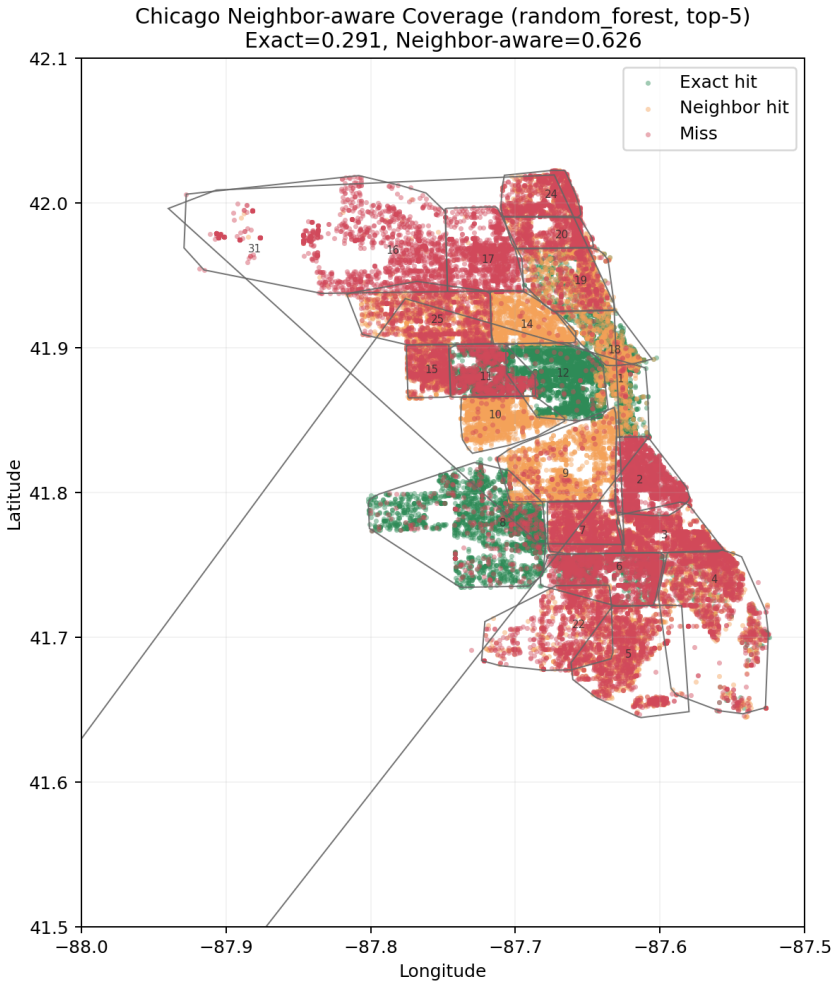

Figure 12: Neighbor-aware geographic event hitmap for Chicago under the 7-day setting.

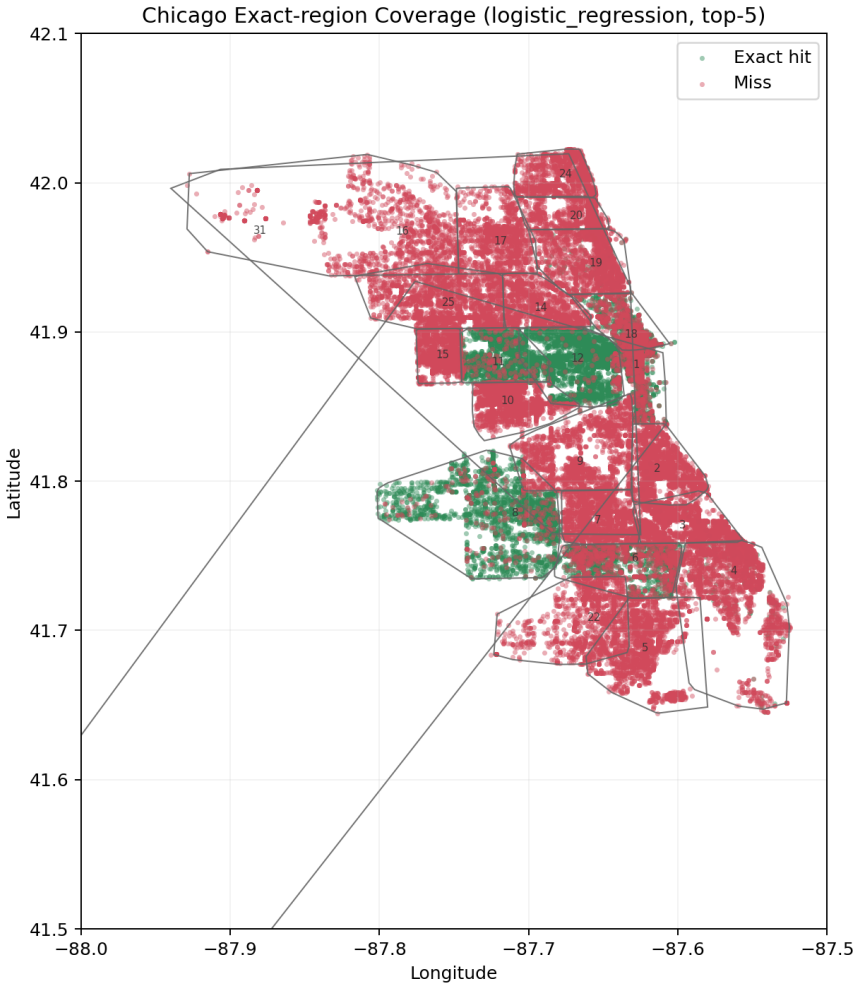

Figure 13: Exact geographic event hitmap for Chicago under the 3-day setting.

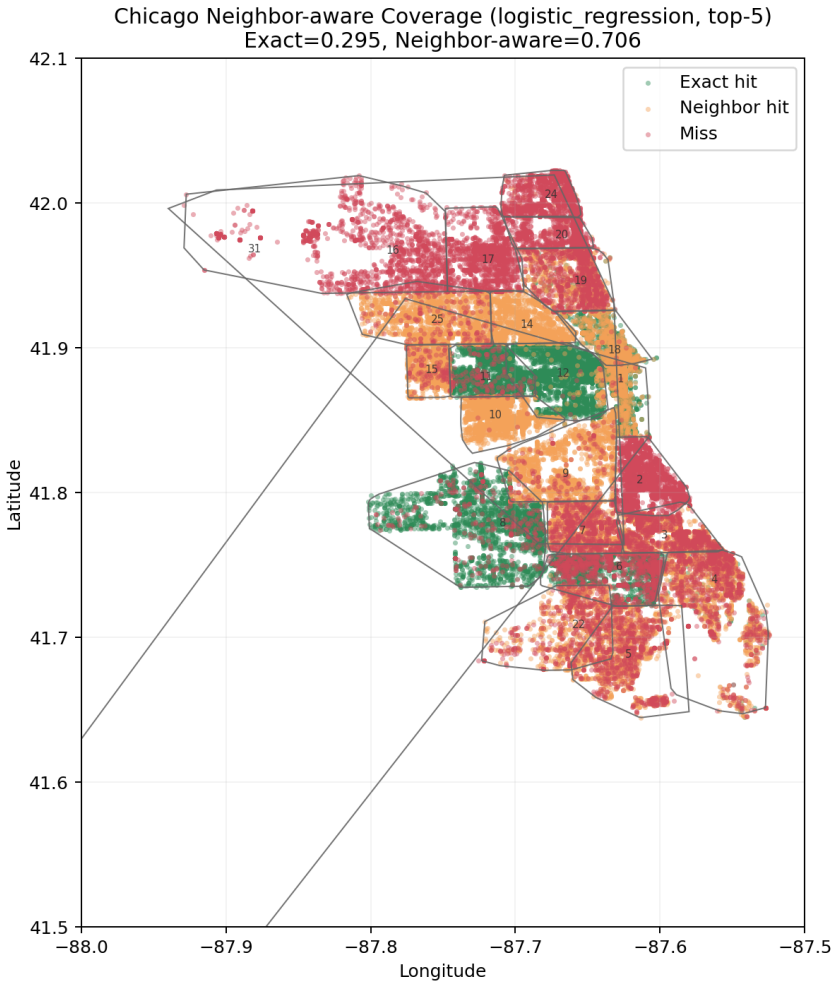

Figure 14: Neighbor-aware geographic event hitmap for Chicago under the 3-day setting.

A.10 Temporal Evaluation Figures

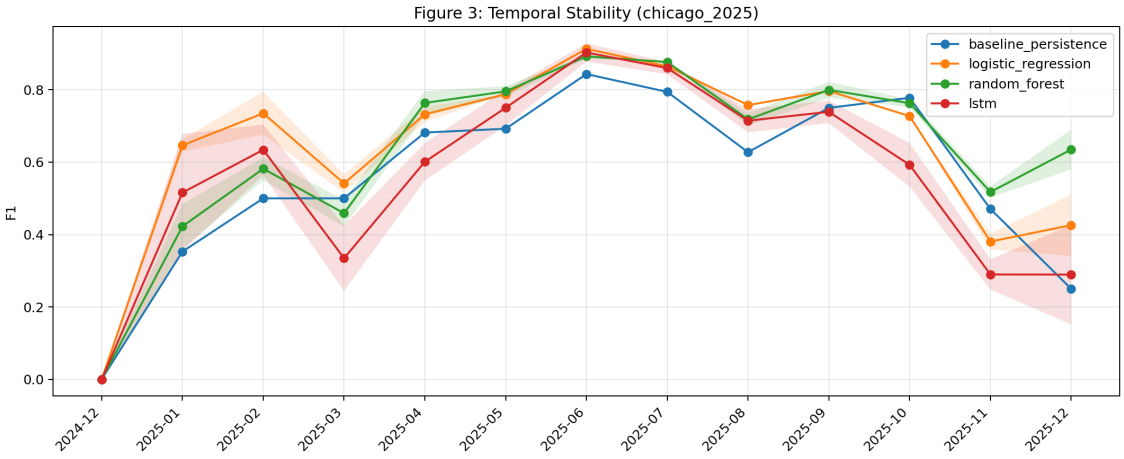

Figure 15: Monthly F1-score trajectories in Chicago 2025 under the 7-day setting.

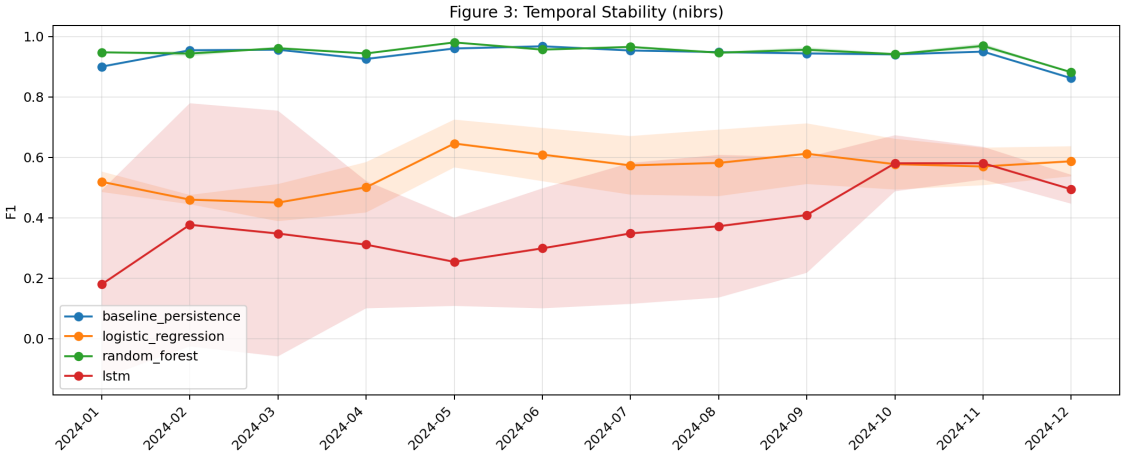

Figure 16: Monthly F1-score trajectories in NIBRS under the 7-day setting.

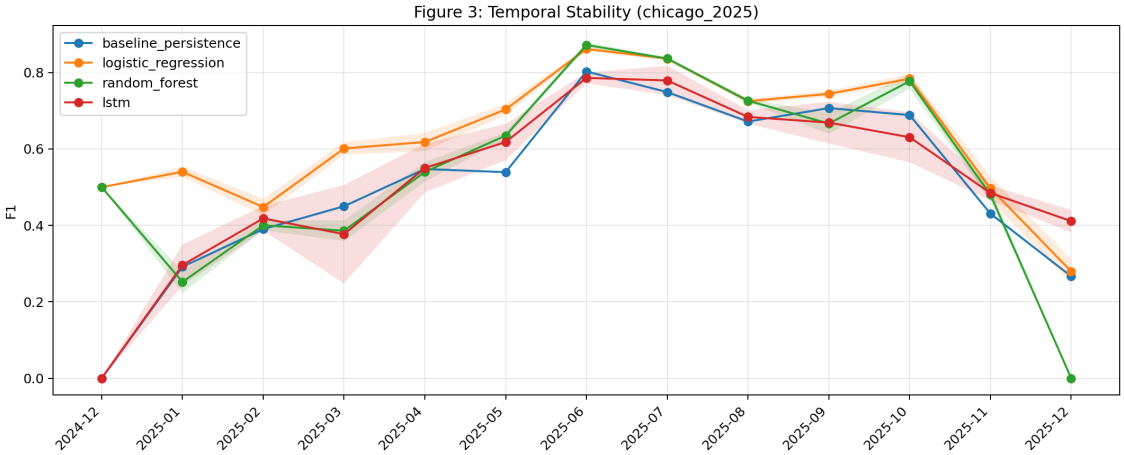

Figure 17: Monthly F1-score trajectories in Chicago 2025 under the 3-day setting.

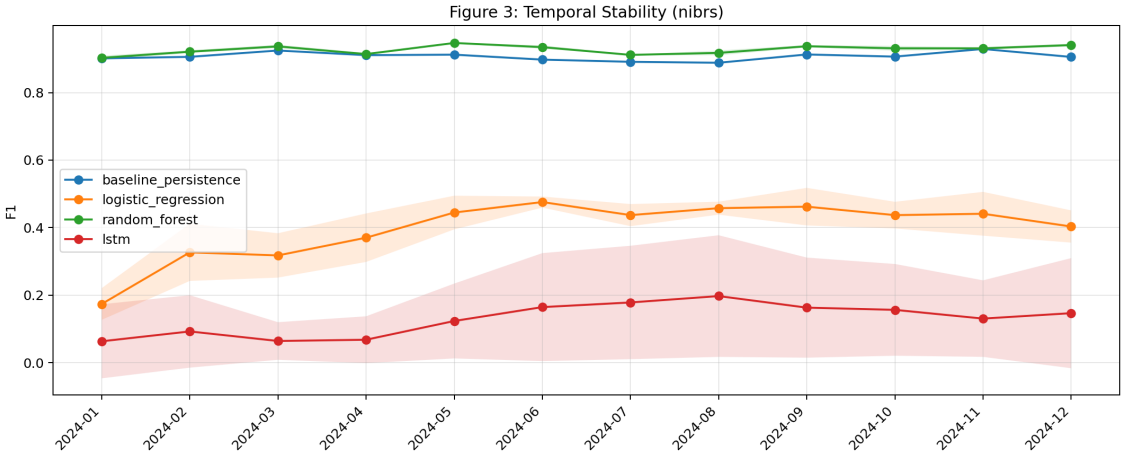

Figure 18: Monthly F1-score trajectories in NIBRS under the 3-day setting.